Master's degree in Electronic Engineering

Politecnico di Torino

Integrated System Architectures
Report LAB2

Group 39

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1 Cvfpu lite simulation

As required by the guidelines document, the original testbench was modified to support the execution of more than ten test vectors. In this work, twelve multiplication operations were selected.

The `data_gen` module was extended to propagate the expected results through a pipeline aligned with the datapath latency. These reference values are displayed in the testbench output, enabling a direct, cycle-by-cycle comparison between the DUT results and the golden model. The test vectors were generated using the *FlexFloat* library and subsequently stored in the `data_gen` ROM used by the testbench.

The test vectors were first expressed in decimal form, as reported in Table 1, while their corresponding *bfloat16* representations, together with the expected results, are shown in Table 2.

The selected test cases include standard arithmetic operations, zero operands, and values close to the limits of quantization and overflow, in order to validate both the functional correctness and the numerical behavior of the bfloat16 multiplication unit.

The *bfloat16* format is a 16-bit floating-point representation. It consists of:

- one sign bit (S),
- eight exponent bits (E) with a bias of 127,
- seven mantissa bits (M) with an implicit leading one.

The numerical value represented by a bfloat16 number is:

$$x = (-1)^S \cdot \left(1 + \frac{M}{2^7}\right) \cdot 2^{(E-127)}$$

The RTL simulation was performed using the provided testbench (`tb_fpnew_top_rtl.sv`) and was executed with the support of two scripts, `sim_rtl.sh` and `sim_rtl.tcl`, which are described in the scripts section of the report.

During RTL simulation, the multiplication involving a negative operand and zero produces a signed zero (-0). The expected results were aligned accordingly in the testbench to enable a bit-accurate comparison. Figure 1 shows a snapshot of the RTL simulation, highlighting the correspondence between the expected bfloat16 results stored in the ROM and the values produced by the multiplier.

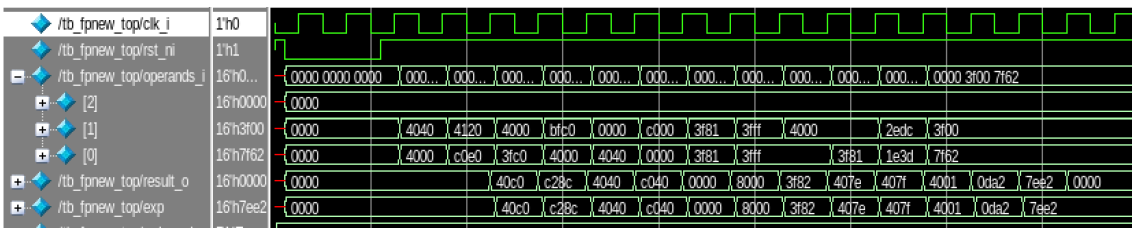

Figure 1: RTL simulation waveform

Idx	Operand A	Operand B	Result
0	2	3	6
1	-7	10	-70
2	1.5	2	3
3	-1.5	2	-3
4	3	0	0
5	-2	0	0
6	1.0078125	1.0078125	1.015625
7	1.9921875	1.9921875	3.96875
8	1.9921875	2	3.984375
9	1.0078125	2	2.015625
10	$1 \cdot 10^{-20}$	$1 \cdot 10^{-10}$	$9.98402 \cdot 10^{-31}$
11	$3 \cdot 10^{38}$	0.5	$1.5023 \cdot 10^{38}$

Table 1: Decimal values of the multiplication test cases

Idx	A (bfloat16)	B (bfloat16)	RES (bfloat16)
0	0_10000000_0000000	0_10000000_1000000	0_10000001_1000000
1	1_10000001_1100000	0_10000010_0100000	1_10000101_0001100
2	0_01111111_1000000	0_10000000_0000000	0_10000000_1000000
3	1_01111111_1000000	0_10000000_0000000	1_10000000_1000000
4	0_10000000_1000000	0_00000000_0000000	0_00000000_0000000
5	1_10000000_0000000	0_00000000_0000000	1_00000000_0000000
6	0_01111111_0000001	0_01111111_0000001	0_01111111_0000010
7	0_01111111_1111111	0_01111111_1111111	0_10000000_1111110
8	0_01111111_1111111	0_10000000_0000000	0_10000000_1111111
9	0_01111111_0000001	0_10000000_0000000	0_10000000_0000001
10	0_00111100_0111110	0_01011101_1011100	0_00011011_0100010
11	0_11111110_1100010	0_01111110_0000000	0_11111101_1100010

Table 2: bfloat16 representations of operands and results

2 Cvfpu lite synthesis

The five synthesis results were obtained through an automated flow based on the scripts `multi_synth.sh` and `synth.tcl`. These scripts implement an iterative synthesis procedure that explores different compilation and optimization strategies and automatically determines the clock period that satisfies the timing constraints.

Once the timing constraints are met, a post-synthesis functional simulation is performed using the netlist testbench `tb_fpnew_top_net.sv` together with the script `sim_net.tcl`. The details of the synthesis and simulation automation are discussed in the scripts section of the report.

Figures 2–6 show the post-synthesis simulation waveforms for the five synthesis strategies considered. The corresponding performance and area results, in terms of

clock period, maximum operating frequency, and silicon area, are summarized in Table 3.

Strategy	**Period [ns]**	**Fmax [MHz]**	**Area [μm^2]**
compile	3.28	304.88	3878.55
compile + optreg	2.28	438.60	3432.46
compile_ultra	2.53	395.26	3351.60
CSA multiplier	2.23	448.43	3613.34
PPARCH multiplier	2.28	438.60	3432.46

Table 3: Post-synthesis performance and area comparison

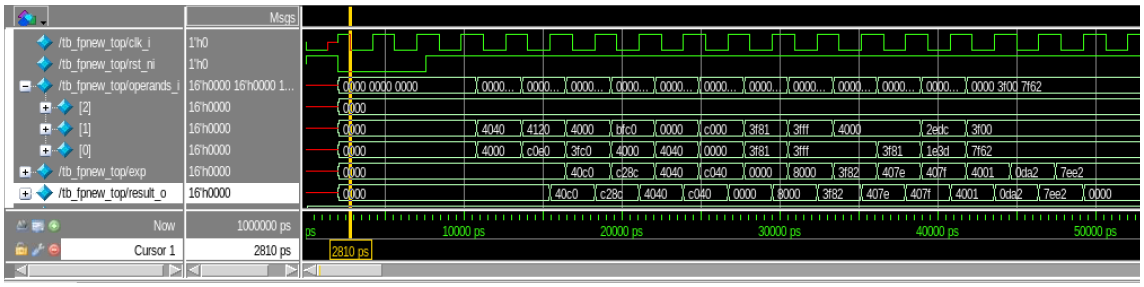

Figure 2: Post-synthesis simulation waveform for the baseline `compile` strategy.

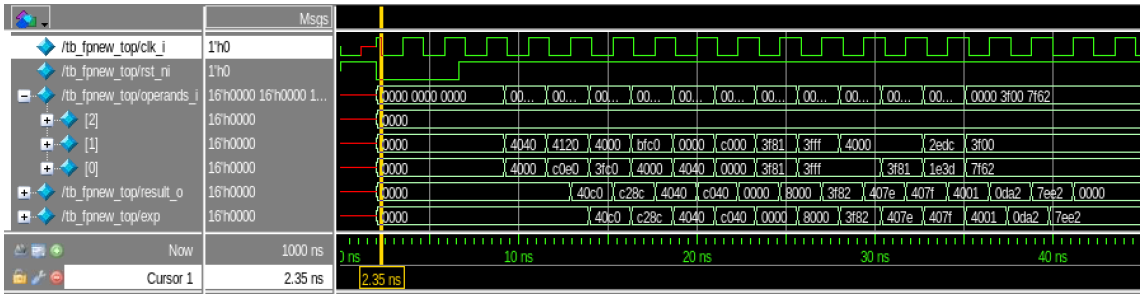

Figure 3: Post-synthesis simulation waveform for the `compile` strategy with register optimization.

The obtained results provide a clear insight into the behavior of the different synthesis strategies considered. The first compilation strategy, which represents the weakest optimization level, yields the worst performance due to limited optimization capabilities and the absence of register retiming. From the `report_resources`, it can be observed that the inferred multiplier adopts a PPARCH architecture optimized for area.

When the `compile ultra` command is used, the performance improves significantly, while also achieving the smallest area among all the considered synthesis strategies.

In the second synthesis mode, where register optimization is enabled through `optimize_registers`, register retiming is applied to redistribute combinational delays across pipeline stages, reducing the critical path while preserving functional

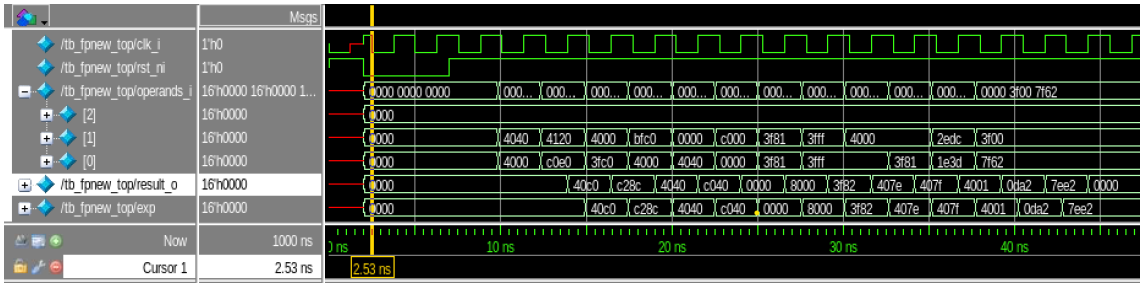

Figure 4: Post-synthesis simulation waveform for the `compile_ultra` synthesis strategy.

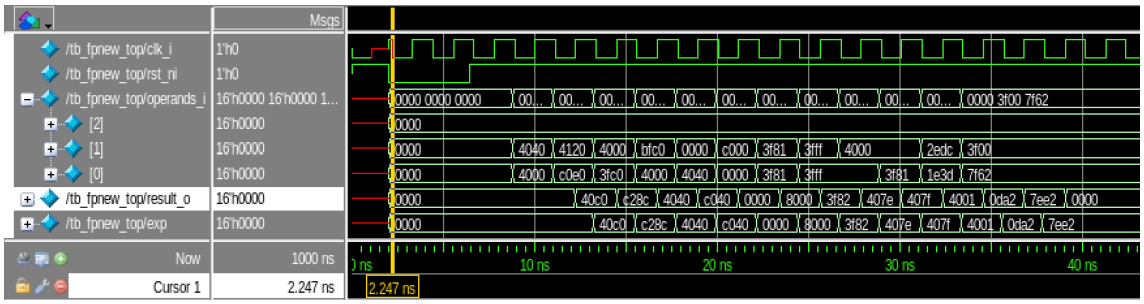

Figure 5: Post-synthesis simulation waveform for the CSA-based FPU.

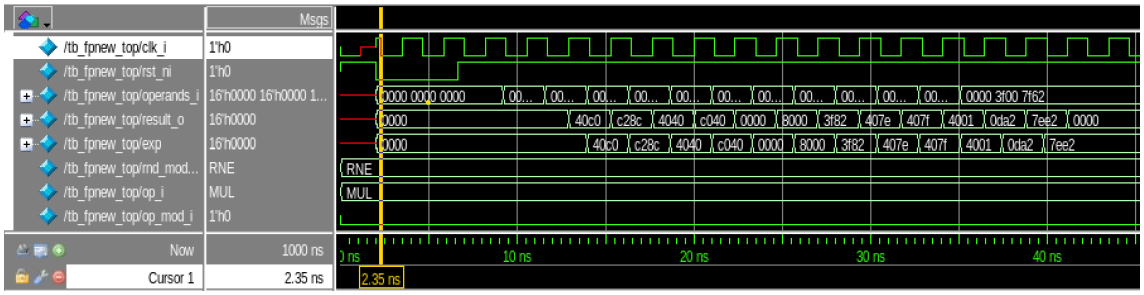

Figure 6: Post-synthesis simulation waveform for the PPARCH-based FPU.

correctness. The obtained results largely overlap with those achieved when enforcing the PPARCH multiplier together with register optimization, since the resource report shows that a PPARCH multiplier is used also in this synthesis.

Finally, it is interesting to note that the synthesis employing the CSA multiplier achieves the best performance in terms of maximum operating frequency, at the cost of an increased area with respect to the other solutions.

3 R4-MBE implementation

In this part of the laboratory, a Radix-4 Modified Booth Encoder (R4-MBE) multiplier has been designed to replace the behavioral implementation of the mantissa multiplier.

The design of the multiplier can be divided into four main blocks:

- the Booth encoder, which converts `operand_b` into Booth-encoded signals;
- the partial product generator, which produces the partial products from the

encoded signals;

- the Wallace tree, which reduces the partial product summation to a two-operand result;
- the carry-propagate adder, which combines the two operands into the final result.

As required by the guidelines, the block diagram of the R4-MBE multiplier is shown in Figure 7.

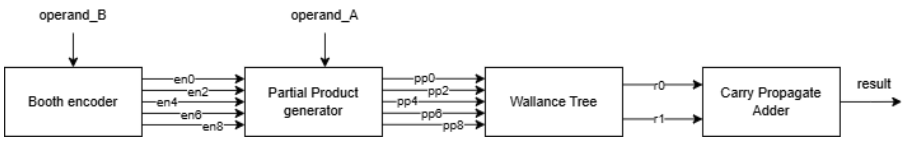

Figure 7: Block diagram of the r4-MBE multiplier.

As required by the guidelines, the designed multiplier takes advantage of a high-radix Booth encoding, which allows the number of partial products to be reduced by half. The Booth encoder takes operand_b as input and produces five encodings. Each encoding is represented in the format (negative, two, zero) [2:0]. The `negative` bit controls the sign of the partial product, the `two` bit selects between a single or doubled multiplicand, and the `zero` bit enables the suppression of the partial product. This representation allows each Booth slice to generate one of the values $\{0, \pm a, \pm 2a\}$. It is worth noting that the last encoding, due to its construction with two leading zeros, can only generate a partial product equal to either a or 0. On the other side the first encoding has been created by adding a 0 in position -1 $b_{-1} = 0$. Following the observation previously done, the last partial product can be represented on 8 bits, since it can only assume the values 0 or a. The Modified Booth encoding scheme employed in this design is summarized in Table 4.

Table 4: Radix-4 Modified Booth encoding

$b_{2j+1}, b_{2j}, b_{2j-1}$	Partial product	neg	two	zero
000	0	0	0	1
001	$+a$	0	0	0
010	$+a$	0	0	0
011	$+2a$	0	1	0
100	$-2a$	1	1	0
101	$-a$	1	0	0
110	$-a$	1	0	0
111	0	0	0	1

The generated encodings are then used to produce the corresponding partial products. In the first step, only the enconding bits *two* and *zero* are used to select between 0, a, and $2a$. Subsequently, an XOR operation with the encoding bit *neg*

is applied to obtain the inverted version of the partial product in the case of $-a$ or $-2a$. The completion of the negation with the addition of 1 in LSB position is handled directly within the Wallace tree in order to reduce the overall critical path.

The Wallace tree has been designed to perform a fast summation of the generated partial products. The need for sign extension when summing negative partial products is addressed using the Bewick sign-extension reduction technique, which significantly reduces the number of required adders in the most significant bit positions, especially for the first partial products. Within the Wallace tree, a combination of half adders and full adders (3:2 compressors) is used to compress the partial products down to two operands. In Figure 8, the structure of the Wallace tree expressed in dot notation can be observed. As discussed previously, based on the construction of the last partial product, it can be represented using 8 bits instead of 9.

The orange squares represent the summation of two elements performed by a half adder, while the green squares represent the summation of three elements performed by a full adder. The purple circles denote the carry signals generated by the summations at the previous level, whereas the green circles represent the corresponding sum outputs. Finally, the black square represents the carry-propagate adder used to compute the final result.

In the Wallace tree, a total of 11 half adders and 27 full adders are used.

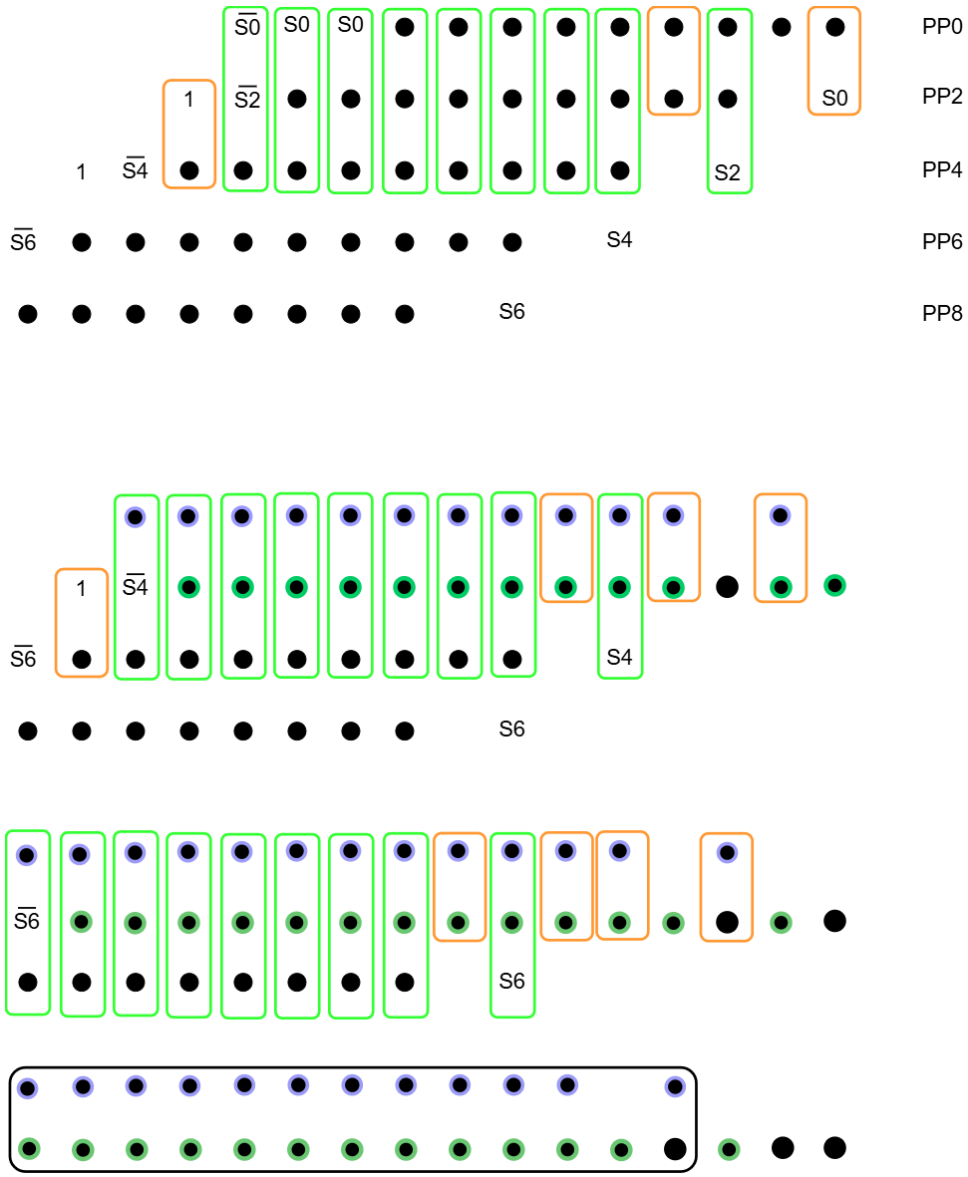

Figure 8: Wallace tree structure expressed in dot notation.

Finally, a behavioral implementation of the carry-propagate adder is used to produce the final result. This part of the multiplier could be further improved by implementing a carry-lookahead adder (CLA), which would further enhance the overall performance of the multiplier.

The obtained design have been tested in a stand-alone configuration. The testbench used is `tb_r4_mbe.sv`, and the corresponding simulation scripts `sim_mymul.sh` and `sim_mymul.tcl` have been developed. This testbench allows a direct visual inspection of the computed results against the expected values. The used test vectors are summarized in Table 5. The selected test vectors include both basic functional cases and corner cases aimed at stressing the Booth recoding, partial product generation, and carry propagation stages.

Figure 9 shows the comparison between the expected and computed multiplication results. Figure 10 highlights the behavior of the radix-4 Booth

Table 5: Test vectors used for the verification of the R4-MBE multiplier

Test	a (bin)	a (dec)	b (bin)	b (dec)	Expected result
0	0000_0000	0	0000_0000	0	0
1	0000_0001	1	0000_0001	1	1
2	0000_0011	3	0000_0101	5	15
3	0000_0111	7	0000_1001	9	63
4	0000_1100	12	0000_0100	4	48
5	0000_1111	15	0000_1111	15	225
6	0000_0011	3	0000_0111	7	21
7	0000_0011	3	0000_1000	8	24
8	1111_1111	255	0000_0011	3	765
9	0000_0001	1	1000_0000	128	128
10	1010_1010	170	0101_0101	85	14450
11	1111_1111	255	1111_1111	255	65025

encoder and the partial product generation, while Figure 11 illustrates the partial summation results at the different levels of the Wallace tree.

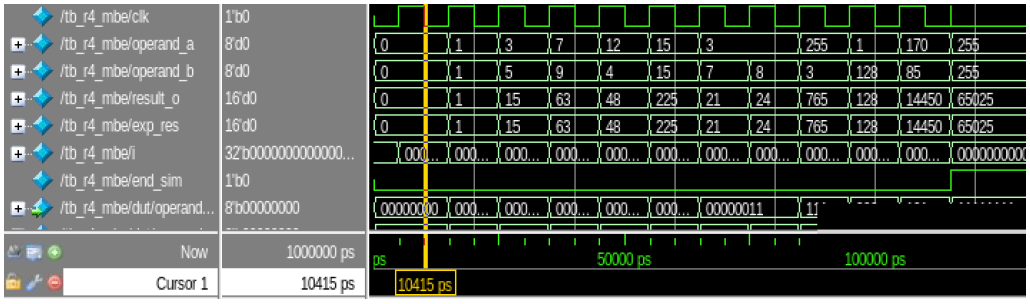

Figure 9: Waveform comparison expected and computed multiplication results for the R4-MBE multiplier.

Finally, the designed multiplier has been integrated into the provided FPU by substituting the original behavioral mantissa multiplier with the newly developed R4-MBE component. An additional simulation has been performed using the adapted scripts `sim_rtl_mymul.sh` and `sim_rtl_mymul.tcl`. The results of this simulation are shown in Figure 12. The test vectors used in this simulation are the same as those presented in the first chapter and are summarized in Table 1 and Table 2.

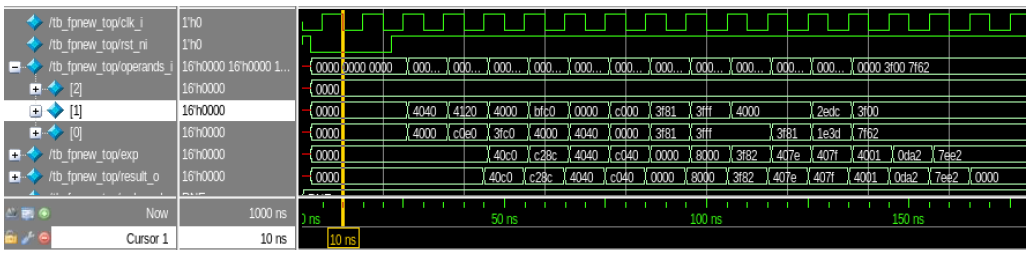

Figure 12: RTL simulation of the FPU after integrating the R4-MBE mantissa multiplier.

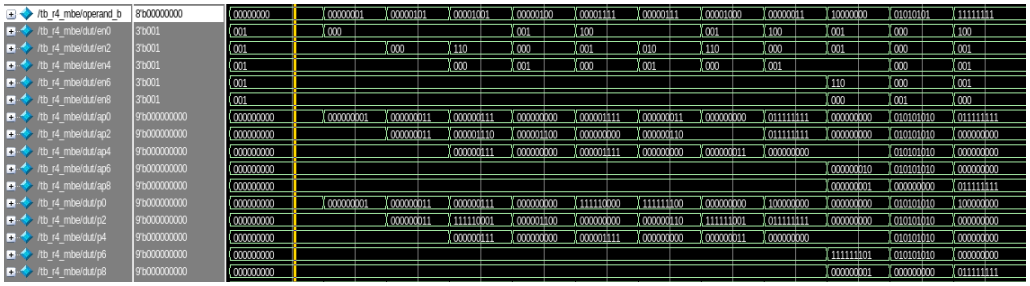

Figure 10: Waveform of the radix-4 Booth encoder and partial product generation for different input patterns.

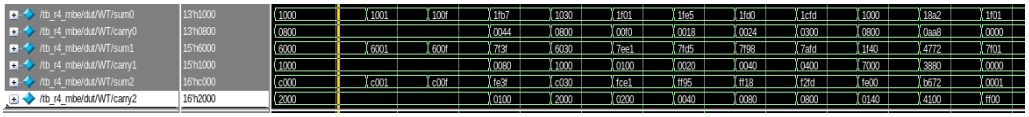

Figure 11: Partial summation results at the different levels of the Wallace tree during the reduction process.

The modified FPU has then been synthesized using the scripts `synth_mymul.sh` and `synth_mymul.tcl`. During synthesis, an automatic search for the maximum operating frequency (*fmax*) of the circuit has been performed. The synthesis results are summarized in Table 6, while a snapshot of the post-synthesis simulation is reported in Figure 13.

The obtained results show only a marginal improvement in terms of maximum operating frequency. However, it is worth noting that the initial `compile ultra` synthesis already employs a PPARCH multiplier. Further performance improvements can be achieved by explicitly using a carry-lookahead adder (CLA) instead of relying on the behavioral description of the carry propagation adder.

On the other hand, an increase in area can be observed. This overhead can be attributed to the fact that the PPARCH implementation used in the first synthesis is already highly optimized for area, and additional architectural optimizations favor speed at the expense of silicon area.

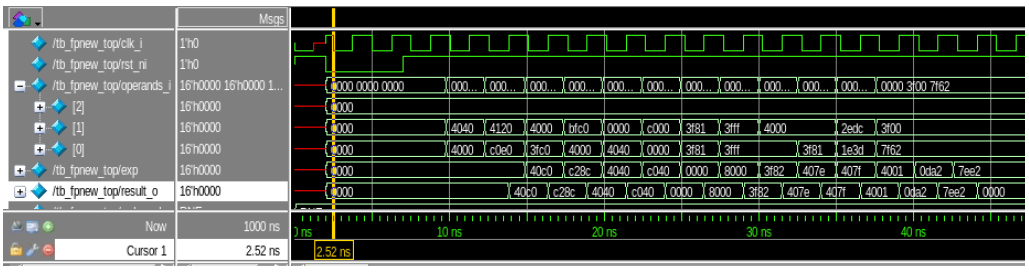

Figure 13: Post-synthesis waveform of the FPU with the integrated R4-MBE multiplier at the new obtained clock.

Table 6: Post-synthesis results of the FPU with the integrated R4-MBE multiplier.

Metric	**Value**
Minimum clock period T_{clk} [ns]	2.52
Total cell area [μm^2]	3696.1
Maximum clock frequency $f_{\max}$ [MHz]	396.8

4 Scripts

Several scripts were developed during the implementation of this laboratory project. All paths used in these scripts are relative; therefore, preserving the original folder structure is essential for correct operation.

Scripts are always created in pairs. Each pair consists of:

- a `.sh` file, responsible for cleaning the working directory and initializing the selected tool;
- a `.tcl` file, responsible for performing operations inside the tool.

In the simulation scripts (`.tcl`), an initial section performs the compilation of all required source files and the testbench. After compilation, the simulation is executed, waveforms are added, and the `run` command is issued.

In the post-synthesis simulation script (`sim_net.tcl`), a different testbench is used (`tb_fpnew_top_net.sv`). In this case, compilation includes the synthesized netlist and the standard cell libraries. This simulation use the environmental variable `CLK_PERIOD` to instantiate the clock at the frequency obtained during synthesis. This is achieved by exploiting the `-g(varname)` flag in the simulation command together with a parametric definition of the clock period in both the testbench and the clock generation files. The usage of this feature is explained in more detail later.

The synthesis script (`synth.tcl`) is derived from previous laboratory instructions. Two environmental variables are used:

- `CLK_PERIOD`, which defines the timing constraint for synthesis;
- `SYNTHMODE`, which selects the compilation and optimization options.

Clock uncertainty, as well as input and output delays, are taken into account. At the end of the process, the script generates the synthesis reports.

The `synth.sh` script allows launching a single synthesis by specifying the desired synthesis mode and timing constraint directly in the file.

The `multi_synth.sh` script was developed to obtain the results of the five synthesis runs required in step 2 of the laboratory.

The core of this script is a Bash algorithm that iteratively performs multiple synthesis runs to determine the maximum achievable circuit frequency. It exploits the tunable clock constraint defined in the synthesis Tcl script to try different clock periods until timing constraints are met. The first attempted value is 1 ns, ensuring

the presence of a violated but constrained path in the timing report. The clock period is then updated based on the slack reported in the synthesis results. This update is performed using a `grep` command combined with `awk` to filter and extract the required value. A new clock period is defined, and a new synthesis is executed. This procedure is repeated until timing constraints are satisfied.

The `multi_synth.sh` script performs this iterative maximum-frequency search for all five synthesis options by leveraging the environmental variable `SYNTHMODE` to set the compilation and optimization flag.

Once the maximum frequency is found, the script automatically launches the post-synthesis simulation using the appropriate clock period, as described earlier.

The `synth_mymul.sh` script performs a similar operation, but only for a single synthesis mode and with the appropriate source files. The post-synthesis simulation is launched at the end of the process.

This set of scripts makes it possible to save several hours of synthesis time and to fully automate a procedure that would otherwise be highly error-prone if performed manually.

SCRITP CITATION WHEN USED COMMENTI SINTESI

A Synthesis with compile

A.1 Timing Report

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 19:49:18 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active
              (rising edge-triggered flip-flop clocked by MY_CLK)
  Endpoint: status_o[UF]
            (output port clocked by MY_CLK)
  Path Group: MY_CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock MY_CLK (rise edge)                                 0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.00       0.00 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.09       0.09 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.00       0.09 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.06       0.15 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
```

```
                                                        0.04       0.19 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.10       0.29 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       0.33 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       0.38 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       0.41 f
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gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.00       0.44 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       0.48 f
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                                                        0.04       0.52 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       0.55 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       0.58 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       0.61 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       0.65 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       0.71 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       0.77 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       0.80 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       0.84 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       0.87 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       0.93 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       0.97 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.09       1.06 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
```

```
                                                        0.04       1.10 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       1.15 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       1.20 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.07       1.27 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       1.33 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       1.36 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       1.42 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       1.49 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       1.52 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       1.56 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       1.61 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       1.65 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       1.69 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       1.75 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       1.80 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       1.85 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       1.89 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.00       1.89 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       1.93 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.02       1.95 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       2.00 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
```

```
                                                        0.03       2.03 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       2.06 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       2.09 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       2.14 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       2.19 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       2.24 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       2.27 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       2.32 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.00       2.32 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       2.36 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.05       2.41 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       2.44 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.03       2.47 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.06       2.53 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.00       2.53 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.04       2.57 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                        0.00       2.57 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_i[4][status][UF] (rr_arb_t
                                                        0.00       2.57 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/U8/ZN (AOI221_X1)
                                                        0.03       2.59 f
gen_operation_groups[0].i_opgroup_block/i_arbiter/U10/ZN (INV_X1)
                                                        0.03       2.62 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_o[status][UF] (rr_arb_tree
                                                        0.00       2.62 r
gen_operation_groups[0].i_opgroup_block/status_o[UF] (fpnew_opgroup_block_0_00000
```

```
                                                        0.00       2.62 r
  i_arbiter/data_i[0][status][UF] (rr_arb_tree_00000004_1_891242)
                                                        0.00       2.62 r
  i_arbiter/U101/ZN (AOI22_X1)                          0.03       2.65 f
  i_arbiter/U102/ZN (NAND2_X1)                          0.04       2.69 r
  i_arbiter/data_o[status][UF] (rr_arb_tree_00000004_1_891242)
                                                        0.00       2.69 r
  status_o[UF] (out)                                    0.02       2.71 r
  data arrival time                                                2.71

  clock MY_CLK (rise edge)                              3.28       3.28
  clock network delay (ideal)                           0.00       3.28
  clock uncertainty                                    -0.07       3.21
  output external delay                                -0.50       2.71
  data required time                                               2.71
  ---------------------------------------------------------------------------
  data required time                                               2.71
  data arrival time                                               -2.71
  ---------------------------------------------------------------------------
  slack (MET)                                                      0.00


1
```

A.2 Area Report

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 19:49:18 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary

Number of ports:                          1986
Number of nets:                           4655
Number of cells:                          2887
Number of combinational cells:            2662
Number of sequential cells:                160
```

```
Number of macros/black boxes:                0
Number of buf/inv:                         627
Number of references:                       15

Combinational area:                3024.686004
Buf/Inv area:                       373.197999
Noncombinational area:              853.860027
Macro/Black Box area:                 0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                   3878.546031
Total area:                 undefined
1
```

B Synthesis with compile + optimize_registers

B.1 Timing Report

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 21:46:14 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: MY_CLK_r_REG84_S1
              (rising edge-triggered flip-flop clocked by MY_CLK)
  Endpoint: MY_CLK_r_REG10_S2
            (rising edge-triggered flip-flop clocked by MY_CLK)
  Path Group: MY_CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock MY_CLK (rise edge)                                 0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  MY_CLK_r_REG84_S1/CK (DFFR_X1)                           0.00       0.00 r
  MY_CLK_r_REG84_S1/Q (DFFR_X1)                            0.11       0.11 r
  U773/ZN (AND2_X1)                                        0.04       0.15 r
  U966/ZN (NAND2_X1)                                       0.02       0.17 f
  U1344/ZN (NAND2_X1)                                      0.04       0.22 r
  U1345/ZN (OAI21_X1)                                      0.04       0.26 f
  U1346/ZN (INV_X1)                                        0.05       0.31 r
  U1362/ZN (NAND2_X1)                                      0.05       0.36 f
  U1375/ZN (INV_X1)                                        0.04       0.40 r
```

```
U1036/ZN (NAND2_X1)                                        0.05       0.45 f
U1376/ZN (OAI22_X1)                                        0.06       0.51 r
U1377/ZN (INV_X1)                                          0.02       0.53 f
U1379/Z (MUX2_X1)                                          0.07       0.61 f
U1403/Z (MUX2_X1)                                          0.08       0.68 f
U1404/ZN (INV_X1)                                          0.04       0.72 r
U1460/ZN (AOI22_X1)                                        0.04       0.76 f
U1461/Z (XOR2_X1)                                          0.08       0.85 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.00       0.85 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.07       0.92 r
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.04       0.95 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.05       1.00 r
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.04       1.04 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.07       1.11 r
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.04       1.15 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.07       1.22 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                           0.00       1.22 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.00       1.22 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.06       1.28 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.04       1.32 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.04       1.36 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.07       1.44 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.06       1.50 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.03       1.53 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.00       1.53 f
```

```
  U1494/ZN (AOI22_X1)                                    0.05       1.58 r
  U1495/ZN (OAI21_X1)                                    0.04       1.62 f
  U2034/ZN (INV_X1)                                      0.04       1.66 r
  U1063/ZN (AND2_X1)                                     0.05       1.71 r
  U1123/ZN (AND3_X1)                                     0.06       1.76 r
  U819/ZN (AND3_X1)                                      0.05       1.82 r
  U811/ZN (AND2_X2)                                      0.05       1.87 r
  U815/ZN (INV_X1)                                       0.03       1.90 f
  U1541/ZN (OAI21_X1)                                    0.06       1.96 r
  U1542/ZN (INV_X1)                                      0.04       2.00 f
  U948/ZN (OR2_X1)                                       0.07       2.06 f
  U864/ZN (AND2_X1)                                      0.04       2.10 f
  U1560/ZN (OAI21_X1)                                    0.04       2.14 r
  U900/ZN (NAND2_X1)                                     0.03       2.17 f
  U1114/ZN (OR2_X1)                                      0.06       2.23 f
  MY_CLK_r_REG10_S2/D (DFFS_X1)                          0.01       2.24 f
  data arrival time                                                 2.24

  clock MY_CLK (rise edge)                               2.35       2.35
  clock network delay (ideal)                            0.00       2.35
  clock uncertainty                                     -0.07       2.28
  MY_CLK_r_REG10_S2/CK (DFFS_X1)                         0.00       2.28 r
  library setup time                                    -0.04       2.24
  data required time                                                2.24
  ---------------------------------------------------------------------------
  data required time                                                2.24
  data arrival time                                                -2.24
  ---------------------------------------------------------------------------
  slack (MET)                                                       0.00


1
```

B.2 Area Report

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 21:46:14 2025
****************************************
```

```
Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary

Number of ports:                          415
Number of nets:                          2602
Number of cells:                         2145
Number of combinational cells:           1884
Number of sequential cells:               246
Number of macros/black boxes:               0
Number of buf/inv:                        465
Number of references:                      45

Combinational area:               2123.743997
Buf/Inv area:                      262.808001
Noncombinational area:            1308.720042
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3432.464039
Total area:                 undefined
1
```

C Synthesis with `compile ultra`

C.1 Timing Report

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 22:04:25 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active
              (rising edge-triggered flip-flop clocked by MY_CLK)
  Endpoint: status_o[UF]
            (output port clocked by MY_CLK)
  Path Group: MY_CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock MY_CLK (rise edge)                                 0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.00       0.00 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.08       0.08 r
  U2467/ZN (AND2_X1)                                       0.05       0.12 r
  U950/ZN (AND4_X2)                                        0.06       0.18 r
  U1750/ZN (NAND4_X1)                                      0.05       0.23 f
  U1766/ZN (AND2_X1)                                       0.05       0.28 f
  U1813/ZN (NAND2_X1)                                      0.03       0.31 r
```

```
U1158/ZN (NAND3_X1)                                    0.03       0.35 f
U1804/ZN (AND2_X1)                                     0.04       0.39 f
U2478/ZN (OAI21_X1)                                    0.05       0.44 r
U2498/ZN (AOI21_X1)                                    0.04       0.47 f
U1903/ZN (NAND2_X1)                                    0.03       0.51 r
U1173/ZN (NAND2_X1)                                    0.05       0.55 f
U1847/ZN (OAI211_X2)                                   0.11       0.66 r
U2565/ZN (AND2_X2)                                     0.09       0.75 r
U2616/ZN (AOI22_X1)                                    0.05       0.80 f
U2618/ZN (AND2_X1)                                     0.04       0.84 f
U951/Z (MUX2_X2)                                       0.07       0.91 f
U1778/ZN (OAI211_X1)                                   0.05       0.96 r
U1085/Z (MUX2_X1)                                      0.05       1.01 r
U1084/ZN (AND2_X2)                                     0.07       1.08 r
U1846/Z (MUX2_X1)                                      0.08       1.16 f
U1862/ZN (NAND2_X1)                                    0.03       1.19 r
U1825/ZN (NAND2_X1)                                    0.03       1.22 f
U1805/ZN (AND3_X1)                                     0.04       1.26 f
U1826/ZN (NAND4_X1)                                    0.03       1.29 r
U1927/ZN (OAI21_X2)                                    0.04       1.34 f
U2879/ZN (OR4_X1)                                      0.14       1.47 f
U1928/ZN (AND2_X1)                                     0.04       1.52 f
U1902/ZN (NAND2_X1)                                    0.03       1.55 r
U1455/ZN (NAND2_X1)                                    0.03       1.58 f
U1872/ZN (AND2_X1)                                     0.04       1.62 f
U1873/ZN (NAND2_X1)                                    0.04       1.66 r
U1875/Z (MUX2_X1)                                      0.09       1.75 f
U1874/ZN (NAND4_X1)                                    0.04       1.78 r
U2947/ZN (OAI21_X1)                                    0.03       1.81 f
U1711/ZN (AND2_X1)                                     0.04       1.85 f
U1786/ZN (OR2_X1)                                      0.05       1.90 f
U2957/ZN (AND2_X1)                                     0.04       1.94 f
status_o[UF] (out)                                     0.02       1.96 f
data arrival time                                                 1.96

clock MY_CLK (rise edge)                               2.53       2.53
clock network delay (ideal)                            0.00       2.53
clock uncertainty                                     -0.07       2.46
output external delay                                 -0.50       1.96
data required time                                                1.96
-------------------------------------------------------------------------
data required time                                                1.96
```

```
  data arrival time                                                  -1.96
  ---------------------------------------------------------------------------
  slack (MET)                          0.00


1
```

C.2 Area Report

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 22:04:25 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary

Number of ports:                          96
Number of nets:                         2823
Number of cells:                        2596
Number of combinational cells:          2463
Number of sequential cells:              132
Number of macros/black boxes:              0
Number of buf/inv:                       421
Number of references:                     44

Combinational area:               2648.827996
Buf/Inv area:                      241.794001
Noncombinational area:             702.772023
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3351.600018
Total area:                 undefined
1
```

D CSA-based Synthesis

D.1 Timing Report

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 22:42:52 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: out_ready_i
              (input port clocked by MY_CLK)
  Endpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_f
            (rising edge-triggered flip-flop clocked by MY_CLK)
  Path Group: MY_CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock MY_CLK (rise edge)                                 0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  input external delay                                     0.50       0.50 r
  out_ready_i (in)                                         0.00       0.50 r
  U847/ZN (OR2_X1)                                         0.03       0.53 r
  U1304/ZN (INV_X1)                                        0.02       0.56 f
  U836/ZN (AND2_X1)                                        0.05       0.60 f
  U2058/ZN (NOR4_X2)                                       0.10       0.71 r
  U1181/ZN (NAND3_X2)                                      0.08       0.78 f
  U1241/ZN (INV_X1)                                        0.08       0.86 r
  U730/Z (CLKBUF_X2)                                       0.09       0.95 r
```

```
U897/Z (CLKBUF_X3)                                          0.09       1.05 r
U1248/Z (MUX2_X1)                                           0.09       1.13 r
U2075/ZN (INV_X1)                                           0.03       1.16 f
U1323/ZN (NAND4_X1)                                         0.04       1.20 r
U1324/ZN (INV_X1)                                           0.03       1.23 f
U1373/ZN (NOR3_X1)                                          0.06       1.29 r
add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
                                                            0.00       1.29 r
add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
                                                            0.03       1.32 f
add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
                                                            0.05       1.37 r
add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
                                                            0.06       1.44 r
add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
                                                            0.00       1.44 r
U1374/ZN (INV_X1)                                           0.03       1.47 f
U1378/ZN (OAI22_X1)                                         0.07       1.54 r
U1121/Z (XOR2_X1)                                           0.07       1.61 r
U1112/ZN (NAND2_X1)                                         0.03       1.64 f
U1425/ZN (INV_X1)                                           0.03       1.67 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.00       1.67 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.02       1.70 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.03       1.73 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.02       1.75 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.06       1.81 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.05       1.86 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.06       1.92 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.04       1.96 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.04       2.00 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                            0.04       2.04 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
```

```
                                                          0.05       2.09 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                          0.02       2.11 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                          0.01       2.12 f
  data arrival time                                                  2.12

  clock MY_CLK (rise edge)                                2.23       2.23
  clock network delay (ideal)                             0.00       2.23
  clock uncertainty                                      -0.07       2.16
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                          0.00       2.16 r
  library setup time                                     -0.04       2.12
  data required time                                                 2.12
  -----------------------------------------------------------------------------
  data required time                                                 2.12
  data arrival time                                                 -2.12
  -----------------------------------------------------------------------------
  slack (MET)                                                        0.00


1
```

D.2 Area Report

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 22:42:52 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary

Number of ports:                          458
Number of nets:                          2772
Number of cells:                         2302
Number of combinational cells:           2024
Number of sequential cells:               258
Number of macros/black boxes:               0
```

```
Number of buf/inv:                        496
Number of references:                      53

Combinational area:               2240.517995
Buf/Inv area:                      286.748001
Noncombinational area:            1372.826044
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3613.344040
Total area:                 undefined
1
```

D.3 Resources Report

```
****************************************
Report : resources
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 22:42:52 2025
****************************************

Resource Sharing Report for design fpnew_top in file ../src/fpnew_top.sv

===============================================================================
|          |              |            | Contained     |                      |
| Resource | Module       | Parameters | Resources     | Contained Operations |
===============================================================================
| r426     | DW01_cmp2    | width=3    |               | gen_operation_groups[0].i_
|               |                      |
|          |              |            |               | gen_operation_groups[0].i_
| r427     | DW01_cmp2    | width=3    |               | gen_operation_groups[1].i_
|               |                      |
|          |              |            |               | gen_operation_groups[1].i_
| r428     | DW01_cmp2    | width=3    |               | gen_operation_groups[2].i_
|               |                      |
|          |              |            |               | gen_operation_groups[2].i_
| r429     | DW01_cmp2    | width=3    |               | gen_operation_groups[3].i_
|               |                      |
|          |              |            |               | gen_operation_groups[3].i_
| r430     | DW01_cmp2    | width=2    |               | i_arbiter/gt_208_G4  |
|          |              |            |               | i_arbiter/lte_209_G4 |
```

```
| r460     | DW01_add     | width=10   |                | gen_operation_groups[0].i_
| r468     | DW01_sub     | width=10   |                | gen_operation_groups[0].i_
| r470     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r472     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r474     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r476     | DW01_sub     | width=5    |                | gen_operation_groups[0].i_
| r478     | DW02_mult    | A_width=8  |                | gen_operation_groups[0].i_
|
|          |              | B_width=8  |                |                       |
| r480     | DW_rightsh   | A_width=36 |                | gen_operation_groups[0].i_
|
|          |              | SH_width=5 |                |                       |
| r482     | DW01_add     | width=29   |                | add_1_root_gen_operation_g
| r484     | DW01_sub     | width=28   |                | gen_operation_groups[0].i_
| r486     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r488     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r494     | DW_cmp       | width=12   |                | gen_operation_groups[0].i_
| r496     | DW01_add     | width=5    |                | gen_operation_groups[0].i_
| r502     | DW01_add     | width=5    |                | gen_operation_groups[0].i_
| r504     | DW_leftsh    | A_width=29 |                | gen_operation_groups[0].i_
|
|          |              | SH_width=5 |                |                       |
| r506     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r508     | DW01_inc     | width=10   |                | gen_operation_groups[0].i_
| r510     | DW01_dec     | width=10   |                | gen_operation_groups[0].i_
| r512     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r514     | DW01_add     | width=15   |                | gen_operation_groups[0].i_
| r1242    | DW01_add     | width=10   |                | add_1_root_add_1_root_gen_
| r1244    | DW01_sub     | width=10   |                | sub_2_root_add_1_root_gen_
| r1246    | DW01_add     | width=10   |                | add_0_root_add_1_root_gen_
| r1966    | DW01_sub     | width=11   |                | sub_1_root_gen_operation_g
| r1968    | DW01_inc     | width=12   |                | add_0_root_gen_operation_g
| r2688    | DW01_sub     | width=10   |                | sub_1_root_gen_operation_g
| r2690    | DW01_inc     | width=10   |                | add_0_root_gen_operation_g
================================================================================


Implementation Report
================================================================================
|                    |                  | Current            | Set            |
| Cell               | Module           | Implementation     | Implementation |
================================================================================
```

```
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
|                     | DW01_inc         | pparch (area,speed)                 |
| add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
|                     | DW01_add         | pparch (area,speed)                 |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
|                     | DW01_sub         | pparch (speed)     |                |
| add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
|                     | DW01_add         | pparch (speed)     |                |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
|                     | DW01_sub         | pparch (area,speed)                 |
| add_0_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
|                     | DW01_inc         | rpl                |                |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
|                     | DW02_mult        | csa                | csa            |
===============================================================================

1
```

E PPARCH-based Synthesis

E.1 Timing Report

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 22:28:00 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: MY_CLK_r_REG84_S1
              (rising edge-triggered flip-flop clocked by MY_CLK)
  Endpoint: MY_CLK_r_REG10_S2
            (rising edge-triggered flip-flop clocked by MY_CLK)
  Path Group: MY_CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock MY_CLK (rise edge)                                 0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  MY_CLK_r_REG84_S1/CK (DFFR_X1)                           0.00       0.00 r
  MY_CLK_r_REG84_S1/Q (DFFR_X1)                            0.11       0.11 r
  U777/ZN (AND2_X1)                                        0.04       0.15 r
  U968/ZN (NAND2_X1)                                       0.02       0.17 f
  U1346/ZN (NAND2_X1)                                      0.04       0.22 r
  U1347/ZN (OAI21_X1)                                      0.04       0.26 f
  U1348/ZN (INV_X1)                                        0.05       0.31 r
  U1364/ZN (NAND2_X1)                                      0.05       0.36 f
  U1377/ZN (INV_X1)                                        0.04       0.40 r
```

```
U1038/ZN (NAND2_X1)                                          0.05       0.45 f
U1378/ZN (OAI22_X1)                                          0.06       0.51 r
U1379/ZN (INV_X1)                                            0.02       0.53 f
U1381/Z (MUX2_X1)                                            0.07       0.61 f
U1405/Z (MUX2_X1)                                            0.08       0.68 f
U1406/ZN (INV_X1)                                            0.04       0.72 r
U1462/ZN (AOI22_X1)                                          0.04       0.76 f
U1463/Z (XOR2_X1)                                            0.08       0.85 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.00       0.85 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.07       0.92 r
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.04       0.95 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.05       1.00 r
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.04       1.04 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.07       1.11 r
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.04       1.15 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.07       1.22 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
                                                             0.00       1.22 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                             0.00       1.22 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                             0.06       1.28 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                             0.04       1.32 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                             0.04       1.36 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                             0.07       1.44 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                             0.06       1.50 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                             0.03       1.53 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                             0.00       1.53 f
```

```
  U1496/ZN (AOI22_X1)                                    0.05       1.58 r
  U1497/ZN (OAI21_X1)                                    0.04       1.62 f
  U2035/ZN (INV_X1)                                      0.04       1.66 r
  U1065/ZN (AND2_X1)                                     0.05       1.71 r
  U1125/ZN (AND3_X1)                                     0.06       1.76 r
  U821/ZN (AND3_X1)                                      0.05       1.82 r
  U813/ZN (AND2_X2)                                      0.05       1.87 r
  U817/ZN (INV_X1)                                       0.03       1.90 f
  U1543/ZN (OAI21_X1)                                    0.06       1.96 r
  U1544/ZN (INV_X1)                                      0.04       2.00 f
  U950/ZN (OR2_X1)                                       0.07       2.06 f
  U866/ZN (AND2_X1)                                      0.04       2.10 f
  U1562/ZN (OAI21_X1)                                    0.04       2.14 r
  U902/ZN (NAND2_X1)                                     0.03       2.17 f
  U1116/ZN (OR2_X1)                                      0.06       2.23 f
  MY_CLK_r_REG10_S2/D (DFFS_X1)                          0.01       2.24 f
  data arrival time                                                 2.24

  clock MY_CLK (rise edge)                               2.35       2.35
  clock network delay (ideal)                            0.00       2.35
  clock uncertainty                                     -0.07       2.28
  MY_CLK_r_REG10_S2/CK (DFFS_X1)                         0.00       2.28 r
  library setup time                                    -0.04       2.24
  data required time                                                2.24
  ---------------------------------------------------------------------------
  data required time                                                2.24
  data arrival time                                                -2.24
  ---------------------------------------------------------------------------
  slack (MET)                                                       0.00


1
```

E.2 Area Report

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 22:28:00 2025
****************************************
```

```
Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary

Number of ports:                          416
Number of nets:                          2603
Number of cells:                         2146
Number of combinational cells:           1884
Number of sequential cells:               246
Number of macros/black boxes:               0
Number of buf/inv:                        465
Number of references:                      45

Combinational area:               2123.743997
Buf/Inv area:                      262.808001
Noncombinational area:            1308.720042
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3432.464039
Total area:                 undefined
1
```

E.3 Resources Report

```
****************************************
Report : resources
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Dec 14 22:28:00 2025
****************************************

Resource Sharing Report for design fpnew_top in file ../src/fpnew_top.sv

===============================================================================
|          |              |            | Contained     |                      |
| Resource | Module       | Parameters | Resources     | Contained Operations |
===============================================================================
| r427     | DW01_cmp2    | width=3    |               | gen_operation_groups[0].i_
|               |                      |
|          |              |            |               | gen_operation_groups[0].i_
| r428     | DW01_cmp2    | width=3    |               | gen_operation_groups[1].i_
```

```
|                |                        |
|          |              |            |                | gen_operation_groups[1].i_
| r429     | DW01_cmp2    | width=3    |                | gen_operation_groups[2].i_
|                |                        |
|          |              |            |                | gen_operation_groups[2].i_
| r430     | DW01_cmp2    | width=3    |                | gen_operation_groups[3].i_
|                |                        |
|          |              |            |                | gen_operation_groups[3].i_
| r431     | DW01_cmp2    | width=2    |                | i_arbiter/gt_208_G4  |
|          |              |            |                | i_arbiter/lte_209_G4 |
| r461     | DW01_add     | width=10   |                | gen_operation_groups[0].i_
| r469     | DW01_sub     | width=10   |                | gen_operation_groups[0].i_
| r471     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r473     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r475     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r477     | DW01_sub     | width=5    |                | gen_operation_groups[0].i_
| r479     | DW02_mult    | A_width=8  |                | gen_operation_groups[0].i_
|
|          |              | B_width=8  |                |                      |
| r481     | DW_rightsh   | A_width=36 |                | gen_operation_groups[0].i_
|
|          |              | SH_width=5 |                |                      |
| r483     | DW01_add     | width=29   |                | add_1_root_gen_operation_g
| r485     | DW01_sub     | width=28   |                | gen_operation_groups[0].i_
| r487     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r489     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r495     | DW_cmp       | width=12   |                | gen_operation_groups[0].i_
| r497     | DW01_add     | width=5    |                | gen_operation_groups[0].i_
| r503     | DW01_add     | width=5    |                | gen_operation_groups[0].i_
| r505     | DW_leftsh    | A_width=29 |                | gen_operation_groups[0].i_
|
|          |              | SH_width=5 |                |                      |
| r507     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r509     | DW01_inc     | width=10   |                | gen_operation_groups[0].i_
| r511     | DW01_dec     | width=10   |                | gen_operation_groups[0].i_
| r513     | DW_cmp       | width=10   |                | gen_operation_groups[0].i_
| r515     | DW01_add     | width=15   |                | gen_operation_groups[0].i_
| r1243    | DW01_add     | width=10   |                | add_1_root_add_1_root_gen_
| r1245    | DW01_sub     | width=10   |                | sub_2_root_add_1_root_gen_
| r1247    | DW01_add     | width=10   |                | add_0_root_add_1_root_gen_
| r1967    | DW01_sub     | width=11   |                | sub_1_root_gen_operation_g
| r1969    | DW01_inc     | width=12   |                | add_0_root_gen_operation_g
```

```
| r2689    | DW01_sub     | width=10   |                | sub_1_root_gen_operation_g
| r2691    | DW01_inc     | width=10   |                | add_0_root_gen_operation_g
================================================================================


Implementation Report
================================================================================
|                    |                  | Current            | Set            |
| Cell               | Module           | Implementation     | Implementation |
================================================================================
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
|                    | DW01_inc         | pparch (area,speed)                 |
| add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
|                    | DW01_add         | pparch (area,speed)                 |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
|                    | DW01_sub         | pparch (speed)     |                |
| add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
|                    | DW01_add         | pparch (speed)     |                |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
|                    | DW01_sub         | pparch (area,speed)                 |
| add_0_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_
|                    | DW01_inc         | rpl                |                |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
|                    | DW02_mult        | pparch (area)      | pparch         |
|                    |                  | mult_arch: and     |                |
================================================================================

1
```

F Synthesis with Custom R4-MBE Multiplier

F.1 Timing Report

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Tue Dec 16 00:21:17 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active
              (rising edge-triggered flip-flop clocked by MY_CLK)
  Endpoint: status_o[UF]
            (output port clocked by MY_CLK)
  Path Group: MY_CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock MY_CLK (rise edge)                                 0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.00       0.00 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].active_format.i_fm
                                                           0.08       0.08 r
  U1255/ZN (AND4_X1)                                       0.07       0.16 r
  U1332/ZN (AND2_X1)                                       0.05       0.20 r
  U1932/ZN (NAND4_X2)                                      0.05       0.26 f
  U1903/ZN (AND2_X1)                                       0.05       0.31 f
  U1968/ZN (NAND2_X1)                                      0.04       0.35 r
```

```
U2847/ZN (OAI211_X1)                                    0.04       0.39 f
U2848/ZN (AND2_X1)                                      0.04       0.44 f
U2860/ZN (OAI21_X1)                                     0.05       0.48 r
U2882/ZN (AOI21_X1)                                     0.03       0.52 f
U2913/ZN (OAI21_X1)                                     0.04       0.56 r
U1792/ZN (NAND2_X1)                                     0.03       0.59 f
U1762/ZN (OR2_X2)                                       0.07       0.66 f
U1761/ZN (OAI211_X2)                                    0.08       0.74 r
U1760/ZN (NOR2_X1)                                      0.04       0.78 f
U2088/ZN (INV_X1)                                       0.06       0.84 r
U3038/ZN (OR2_X1)                                       0.05       0.89 r
U3039/ZN (NAND4_X1)                                     0.05       0.93 f
U3040/ZN (NAND2_X1)                                     0.03       0.96 r
U3041/ZN (NAND4_X1)                                     0.04       1.00 f
U3042/ZN (OAI211_X1)                                    0.04       1.04 r
U1503/ZN (AND3_X1)                                      0.07       1.11 r
U1991/ZN (INV_X1)                                       0.03       1.13 f
U1275/ZN (NAND2_X1)                                     0.03       1.16 r
U1273/Z (BUF_X1)                                        0.04       1.20 r
U2050/ZN (INV_X1)                                       0.02       1.23 f
U2018/ZN (AOI21_X1)                                     0.05       1.27 r
U1891/ZN (OR2_X1)                                       0.04       1.31 r
U1868/ZN (AND2_X1)                                      0.05       1.36 r
U1900/ZN (NAND4_X1)                                     0.05       1.41 f
U1993/ZN (NAND2_X1)                                     0.05       1.46 r
U1269/ZN (OR2_X1)                                       0.04       1.50 r
U2056/ZN (NAND2_X1)                                     0.03       1.53 f
U2055/ZN (NAND3_X1)                                     0.03       1.56 r
U1912/ZN (AND2_X1)                                      0.05       1.61 r
U1268/ZN (AND3_X1)                                      0.06       1.66 r
U1184/ZN (AND2_X2)                                      0.05       1.71 r
U1995/ZN (OAI21_X1)                                     0.03       1.74 f
U2023/ZN (INV_X1)                                       0.02       1.76 r
U1812/ZN (AND3_X1)                                      0.05       1.81 r
U2010/ZN (NAND3_X1)                                     0.04       1.85 f
U2006/ZN (NAND2_X1)                                     0.03       1.88 r
U1998/ZN (AND2_X1)                                      0.05       1.93 r
status_o[UF] (out)                                      0.02       1.95 r
data arrival time                                                  1.95

clock MY_CLK (rise edge)                                2.52       2.52
clock network delay (ideal)                             0.00       2.52
```

```
  clock uncertainty                                        -0.07       2.45
  output external delay                                    -0.50       1.95
  data required time                                                   1.95
  -------------------------------------------------------------------------
  data required time                                                   1.95
  data arrival time                                                   -1.95
  -------------------------------------------------------------------------
  slack (MET)                                                          0.00


1
```

F.2 Area Report

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Tue Dec 16 00:21:17 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary

Number of ports:                          96
Number of nets:                         3087
Number of cells:                        2884
Number of combinational cells:          2751
Number of sequential cells:              132
Number of macros/black boxes:              0
Number of buf/inv:                       484
Number of references:                     44

Combinational area:               2803.107994
Buf/Inv area:                      271.054002
Noncombinational area:             702.772023
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3505.880017
Total area:                 undefined
```

```
1
```